Wireless networking

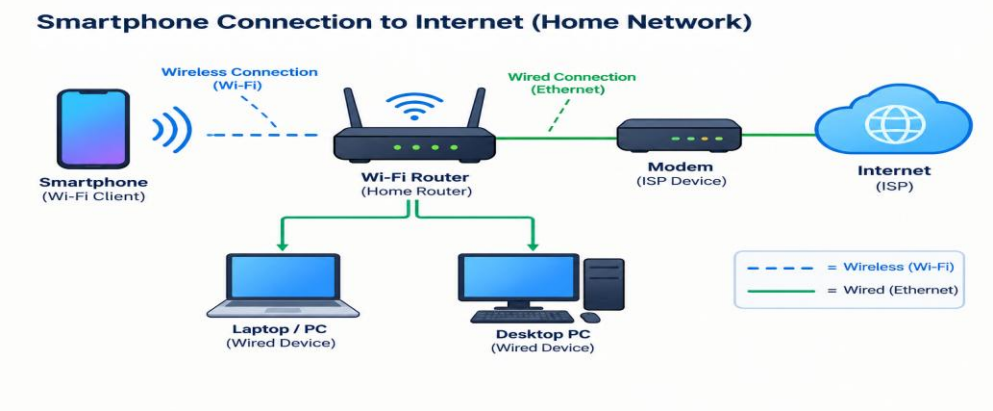
1.List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

ANS:

Wi-Fi Standard	Frequency Band	Key Difference
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Speed up to 600 Mbps.
802.11ac (Wi-Fi 5)	5 GHz	Faster speeds, up to about 6.9 Gbps.
802.11ax (Wi-Fi 6)	2.4 GHz and 5 GHz	Better performance with many devices connected, with speeds up to about 9.6 Gbps.

2.Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.

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3.Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

Hint: Consider both signal strength and security when choosing.

ANS:

Practical: Scan Available Wi-Fi Networks

No.	SSID (Wi-Fi Name)	Signal Strength	Security Type
1	Home_WiFi	Excellent (90%)	WPA2-Personal
2	JioFiber_5G	Good (75%)	WPA3-Personal
3	Guest_WiFi	Weak (35%)	Open (No Password)

Which Wi-Fi Network Would I Choose?

I would choose JioFiber_5G because it has good signal strength and uses WPA3 security. WPA3 provides stronger protection against unauthorized access than older security methods. A strong signal also helps provide a stable internet connection.

Steps to Scan Wi-Fi Networks

1. Open Settings on your laptop or smartphone.
2. Go to Wi-Fi settings.
3. Turn on Wi-Fi.
4. Check the available network names (SSID).
5. Note the signal strength and security type, if displayed.
6. Record the details of at least three networks in a table.

4.Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

ANS:

SSID Hiding in Wi-Fi Networks

Meaning: SSID hiding means hiding the name of a Wi-Fi network so that it does not normally appear in the list of available Wi-Fi networks on nearby devices. Users may need to enter the Wi-Fi name manually to connect.

1. Advantage: It makes the Wi-Fi network less visible to casual users because its name is not displayed in the normal Wi-Fi list.

2. Disadvantage: It does not provide strong security. Someone can still discover a hidden network using network-scanning tools, so a strong password and WPA2/WPA3 security are still necessary.